

Name

Saghir Ali

Sap-ID

67005

Quiz

(02) Algorithm

Dept

AI 4-1

Date

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Q:- 1

Given Recurrence

$$T(n) = 2T(n/2) + n$$

we will use substitution

our induction goal is

$$T(n) \leq cn \quad \& \quad n \geq n_0$$

$$T(n) = O(n \lg n)$$

$$T(n/2) \leq cn/2 \lg(n/2)$$

Proof:-

$$T(n) = 2T(n/2) + n \leq 2c(n/2) \lg(n/2) + n$$



$$-cn \lg n - cn + n \leq cn \lg n$$

$$\text{if: } -cn + n \leq 0$$

$$\Rightarrow c \geq 1$$

The above was iterative method now we expand the Recurrence

$$T(n) = 2[2T(n/4) + n/2] + n$$

$$= 4T(n/4) + n + n$$

$$= 4[2T(n/8) + n/4] + 2n$$

$$= 8T(n/8) + 3n$$

As we have  $n/8$  so  $\log$  is necessary  
so

$$T(n) = nT(1) + n \log n$$

## Time Complexity:

By using above iterative method

$$T(n) \leq cn \log n - (c-1)n$$

So for large  $c$

$$T(n) = O(n \log n)$$

Time Complexity is

$$T(n) = O(n \log n)$$



~~Q: 2 B~~

Given Recurrence :-

$$T(n) = T(n-1) + \log n$$

Expanding :-

$$T(n) = T(n-1) + \log n$$

$$T(n) = T(n-2) + \log(n-1) + \log n$$

$$T(n) = T(n-3) + \log(n-2) + \log(n-1) + \log n$$

we will use this ~~until~~  
until  $T(1)$

$$T(n) = T(1) + \sum_{i=2}^n \log i$$

$$\text{As } \log 1 + \log 2 + \log 3 + \dots + \log n$$

$$\Rightarrow \log(n!)$$

So if we approximate  
the Recurrence

$$\log(n!) \approx n \log n$$



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Time complexity:-

$$T(n) = O(n \log n)$$

So the time complexity  
is  $O(n \log n)$